Still Citing the Old Law? Statutory Knowledge and Temporal Anchoring in Thai Legal Question Answering

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Abstract

People who cannot afford a lawyer increasingly ask language models about the law. We ask whether those models know Thai statutory law, and whether they keep answering with rules that have since been repealed. We build a bilingual benchmark of 60 synthetic vignettes covering four dated amendments to Thai law between eight months and five years old. Fourteen of the vignettes test rules that did not change over that period. We evaluate nine models across two prompt languages with three runs each (3,036 responses). The clearest result is not about staleness. On the unchanged rules the most recent frontier models answer 93–100% correctly while a Thai-specific model answers 31–41% and a Chinese open model 64–66%: a gap of roughly sixty percentage points, though the two weaker models are also the smaller ones, so training and capability are not separated here. Anchoring to repealed rules is a real but smaller effect, concentrated in older model versions, and it declines in five of the six old/new model pairs we test. Two hypotheses we set out to test do not hold: anchoring does not fade monotonically with amendment age, and Thai prompts are not uniformly worse than English. Across all wrong answers, 97.7% contain no expression of uncertainty.

1 Introduction

Thailand has roughly seventy million Thai speakers and a legal-aid system that does not reach most of them. When the law changes and a language model is the first place someone looks, the cost of a stale answer falls on exactly the users with the least ability to check it against the Gazette.

We ask two questions. Do language models know Thai statutory law at all? And when a provision is amended, do they keep answering with the superseded version? These are different failures with different remedies: a model that knows the law but is a year out of date can be fixed with retrieval; a model that does not know the law is a different problem. We built the benchmark so the two can be told apart. Alongside items testing amended provisions, it includes control items testing rules that did not change over the same period. A model that answers the controls correctly and the amended provisions incorrectly is anchored to old law. A model that fails both simply does not know it.

We operationalise this as five research questions. **RQ1** Do models correctly state rules that have *not* changed? **RQ2** How often do they answer with a repealed rule? **RQ3** Does anchoring decline in newer model versions, and fade as an amendment ages? **RQ4** Are Thai-language prompts answered less accurately than English ones about the same Thai law? **RQ5** Do models signal uncertainty when wrong?

Our contributions are a bilingual Thai/English benchmark of 60 vignettes across four dated amendments with every gold label traceable to a Royal Gazette citation; an evaluation of nine models

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: GlobalSouthAI.

Table 1: The four amendments, with Royal Gazette citations. Ages relative to August 2026. CCC = Civil and Commercial Code; LPA = Labour Protection Act.

Group	Instrument (Gazette)	Change	In force
A	Emergency Decree amending the CCC, B.E. 2564 (vol. 138, pt. 26a, p. 1)	s.7: 7.5% → 3%/yr s.224: 7.5% → 5%/yr	11 Apr 2021 (5 yr 4 mo)
B	Act amending the CCC (No. 23), B.E. 2565 (vol. 139, pt. 69a)	s.1097: 3 → 2 promoters	7 Feb 2023 (3 yr 6 mo)
C	Act amending the CCC (No. 25), B.E. 2568 (vol. 142, pt. 14a)	s.1567(2): "as appropriate" → cruelty/violence limits	25 Mar 2025 (17 mo)
D	LPA (No. 9), B.E. 2568 (vol. 142, pt. 74a, p. 41)	s.41: 98 → 120 days s.59: 45 → 60 paid days	7 Dec 2025 (8 mo)

spanning three Western providers, one Chinese open model and one Thai-specific model; a design that separates anchoring from absence of knowledge and from retroactive application of new law; and a full public record of the grading rubric's development, including three items we excluded and the errors we found in our own scorer. Related work is discussed in Appendix A.

2 Benchmark

Amendments. We selected four amendments to Thai law that are dated, numerically or textually unambiguous, and politically uncontroversial [Emergency Decree, 2021, CCC No. 23, 2022, CCC No. 25, 2025, LPA No. 9, 2025] (Table 1). The spread of ages lets us ask whether anchoring decays as an amendment recedes. Group D has a further property we exploit: maternity leave has been amended twice — 90 days under the original Labour Protection Act B.E. 2541 [LPA, 1998], 98 under Act No. 7 B.E. 2562 [LPA No. 7, 2019], and 120 today — so a model can be anchored one amendment back or two, and we record which.

Vignettes. We wrote 60 short fact patterns, each in matched Thai and English. They are synthetic rather than drawn from decided cases: we gave up some realism so that the benchmark can be shared freely and holds no personal data. Each vignette carries a *temporal frame* (post-frame items date the facts after the amendment so the current rule is correct; pre-frame items date them before, so the superseded rule is correct — without this, a wrong answer cannot be told from ignorance), a *probe type* (knowledge items ask what the rule is; application items give facts requiring it to be used), and a role: fourteen *control* items test rules that did not change at any point across the period the four amendments span, and three *boundary* items straddle an amendment date or present a matched pair of facts either side of one. Every gold label traces to the text of an amending instrument in the Royal Gazette; where a superseded rule was needed, we obtained the earlier instrument too.

Effective dates. Groups A, B and C state their commencement unambiguously. Group D reads "upon the expiry of thirty days from the date of publication", which yields 8 December 2025 under the Civil Code's general counting rule and 7 December under practitioner usage; we adopt 7 December and keep group D vignettes three months clear of it, so that a "98 days" response cannot be confounded with a day-count disagreement. Boundary items are placed in groups A and B, whose dates are exact.

3 Setup

We evaluate `gpt-4o-2024-11-20`, `gpt-5-2025-08-07`, `gpt-5.5-2026-04-23`, `claude-sonnet-4.5`, `claude-sonnet-5`, `gemini-2.5-pro`, `gemini-3.1-pro-preview`, `qwen3-235b-a22b-2507` and `typhoon-v2.5-30b-a3b-instruct`; the last is trained specifically for Thai. Version strings are those the API returned. Each vignette is asked in both languages, three times, at temperature 0 where the model accepts it, with a fixed system prompt and no tools or retrieval: 3,036 responses (a full grid would be 3,240; `gpt-5` was collected only partially). Collection and grading are separate programs, so the rubric was revised over seven pilot rounds without re-querying. Every change, with the benchmark, raw outputs, grader and analysis code, is released at `https://github.com/EarthBBMP/thai-legal-temporal-grounding`; the net effect of all revisions was to *reduce* measured anchoring.

Table 2: Control accuracy — rules that did not change. Wilson 95% intervals. `gpt-5` has a much smaller denominator and is shown for completeness.

Model	Thai	English
gpt-5.5	100.0% [91.6–100.0] (42/42)	100.0% [91.6–100.0] (42/42)
gemini-3.1	100.0% [91.6–100.0] (42/42)	95.2% [84.2–98.7] (40/42)
gemini-2.5	100.0% [91.6–100.0] (42/42)	92.7% [80.6–97.5] (38/41)
claude-5	95.2% [84.2–98.7] (40/42)	92.9% [81.0–97.5] (39/42)
claude-4.5	92.5% [80.1–97.4] (37/40)	85.7% [72.2–93.3] (36/42)
gpt-4o	75.6% [60.7–86.2] (31/41)	90.5% [77.9–96.2] (38/42)
qwen3	65.9% [50.5–78.4] (27/41)	64.3% [49.2–77.0] (27/42)
typhoon-2.5	40.5% [26.3–56.5] (15/37)	31.0% [19.1–46.0] (13/42)
gpt-5*	85.7% [48.7–97.4] (6/7)	88.9% [56.5–98.0] (8/9)

Table 3: RSCR by old/new model pair. Negative differences mean the newer model cited the repealed rule less often. No significance test is applied; differences are from unrounded rates.

Pair	Lang.	Older	Newer	Δ (pp)
gpt-4o → gpt-5.5	th	26.8% [17.9–38.1]	7.6% [3.5–15.6]	−19.2
gpt-4o → gpt-5.5	en	23.1% [15.1–33.6]	14.8% [8.7–24.1]	−8.3
claude-4.5 → claude-5	th	17.3% [10.4–27.4]	10.3% [5.3–19.0]	−7.1
claude-4.5 → claude-5	en	40.0% [29.7–51.3]	13.9% [8.0–23.2]	−26.1
gemini-2.5 → gemini-3.1	th	20.8% [13.2–31.1]	17.1% [10.3–27.1]	−3.7
gemini-2.5 → gemini-3.1	en	13.3% [7.4–22.8]	14.8% [8.7–24.1]	+1.5

Responses are labelled `correct`, `repealed` (states or applies the superseded rule), `over_applied` (applies the current rule to earlier facts — a distinct failure we keep separate), `wrong_other`, `refusal` or `ambiguous`, with flags for hedging and for quoting wording deleted from the statute. All proportions carry Wilson 95% intervals. RSCR, our headline anchoring measure, is the share of scorable post-frame responses labelled `repealed`. Three vignettes are excluded from every metric for reasons recorded in Appendix B, leaving 2,881 responses in the analysis set.

Training cutoffs. Groups A (2021) and B (2023) predate every model's training data; groups C (March 2025) and D (December 2025) do not. Pair comparisons should be read primarily on A and B, and the C and D rates as an upper bound that mixes anchoring with simple absence of information.

4 Results

In brief: (1) statutory knowledge, not staleness, separates the models — the control gap is roughly sixty points (RQ1); (2) anchoring to repealed rules is real but smaller, concentrated in older versions (RQ2); (3) it declines in five of six old/new pairs but does *not* fade monotonically with amendment age (RQ3); (4) the Thai/English gap has no consistent direction (RQ4); (5) 97.7% of wrong answers carry no expression of uncertainty (RQ5).

Knowledge, not staleness, is the dominant effect. Table 2 gives accuracy on rules that did not change. The Thai-specific model has the lowest point estimate in both languages; its intervals do not overlap those of any Western model except `gpt-5`, whose partial denominator leaves its interval uninformative, and against the Chinese open model the separation holds in English while the Thai intervals overlap. This constrains everything that follows: for `typhoon-2.5` and `qwen3` we cannot claim their errors are about amendment, because they get the unchanged rules wrong too. The anchoring analysis applies cleanly only to models whose control accuracy is high.

Anchoring is real and mostly declines with recency. Five of six pairs move in the expected direction (Table 3); the two largest movements are also the ones with least interval overlap. Group C supplies the cleanest single piece of evidence: the 2025 amendment replaced the phrase *tam samkhuan* ("as appropriate") with an explicit prohibition on cruelty and violence, and 30.8% [25.3–

36.9] (74/240) of group C responses in both languages still quote the deleted phrase — a lower bound, since it can only appear naturally in Thai answers.

Two hypotheses do not hold. Group B, in force three and a half years, has the *lowest* RSCR of the four groups (3.7% [1.6–8.4] Thai; 10.4% [6.3–16.8] English), lower than group A at five years; we claim no gradient. And where Rohera et al. [2025] find multilingual models more accurate in English even on Indic-context questions, we do not reproduce that for Thai law: `claude-4.5` is markedly worse in English (40.0% vs 17.3%), as is `qwen3` (40.8% vs 30.7%), while `gemini-2.5` goes the other way.

Application is harder than recall, and errors arrive without hedging. Pooled post-frame RSCR is 18.1% on knowledge items against 26.1% on application items in Thai, and 20.1% against 29.4% in English; the intervals overlap and we report a direction, not a separated effect. Of 799 wrong responses, 781 contain no expression of uncertainty: 97.7% [96.5–98.6]. This is the behaviour Dahl et al. [2024] describe for US case law — models failing to anticipate their own legal hallucinations — reproduced for statutory rules in a different jurisdiction and language.

Validation. Hand-checking every response to the three boundary items gave 87.2% [81.0–91.5] (136/156) agreement with the automatic label; 11 of the 20 disagreements were labels calling a wrong response correct, so reported anchoring rates are, if anything, slight underestimates. Reproducibility observations on reasoning models are in Appendix B.

What this means for the Global South

The population most likely to ask a model about Thai labour or civil law is the population least able to verify the answer. Three things follow from our data.

The one Thai-specific model we tested is the least accurate on Thai statutes in our set, including on rules that did not change. One model is not a class, and it is also among the smallest we tested, so we cannot attribute the gap to its training rather than its capacity. But the developers of the strongest open Thai legal model report the same shape of result — far higher accuracy with the relevant sections supplied than from memory [iApp Technology, 2026] — and retrieval rather than scale is the remedy both point towards, consistent with Akarajaradwong et al. [2025] for Thai and with Dahl et al. [2024]'s caution against unsupervised deployment.

Access is itself a finding. No Thai-language model is offered through the largest aggregate inference platform we surveyed, across 416 models; reaching one required a provider-specific endpoint. A researcher or deployer in Thailand starts from a narrower set of options than the same person working in English, and the options that are marketed for the language are the ones that performed worst here.

The absence of hedging is a deployment risk on its own. A wrong answer delivered with visible uncertainty invites checking; 97.7% of the wrong answers here do not. We report the negative and non-monotonic results alongside the positive ones because a benchmark that only reports what it hoped to find is not a safety instrument. Nothing in the design is specific to Thai — a dated gazette, a few unambiguous amendments, and unchanged rules for controls are all it needs — and we would like to see it built for the languages next to ours.

Limitations

Vignettes are synthetic, so construct validity is limited; four amendments in one jurisdiction is a narrow base; the task is statute identification, not outcome forecasting [Medvedeva et al., 2023]. The two lowest-scoring models are also the two smallest by active parameters, and we do not know the sizes of the proprietary models, so the control gap confounds training with capability. Some control provisions were last amended in 2019; they are controls for the 2021–2025 window, not provisions untouched since enactment. Grader agreement was measured on the three boundary items only. Intervals treat three runs of an item as independent, which they are not. Three vignettes were excluded after inspecting pilot data, using criteria formed at that point. One model refused a substantial share of group C items while no other model refused any. Fourteen responses were re-collected with a higher token budget after returning empty.

References

P. Akarajaradwong, P. Pothavorn, C. Chaksangchaichot, P. Tasawong, T. Nopparatbundit, and S. Nutanong. NitiBench: A comprehensive study of LLM framework capabilities for Thai legal question answering. *arXiv:2502.10868*, 2025. https://arxiv.org/abs/2502.10868.

O. Atukorala, R. Appuhami, and N. de Silva. LawChain: A resource-efficient blueprint for legal information retrieval in low-resource jurisdictions. *GlobalSouthML Affinity Event, ICML*, 2026. https://icml.cc/virtual/2026/78334.

M. Dahl, V. Magesh, M. Suzgun, and D. E. Ho. Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis*, 16(1):64–93, 2024. doi:10.1093/jla/laae003.

iApp Technology and AIEAT. OpenThai 2.0 Legal ThaiLLM (Nemotron-3-Nano-30B-A3B), model card. https://huggingface.co/iapp/openthai2.0-legal-thaillm-nemotron-3-nano-30b-a3b, 2026.

M. Medvedeva, M. Wieling, and M. Vols. Rethinking the field of automatic prediction of court decisions. *Artificial Intelligence and Law*, 31(1):195–212, 2023. doi:10.1007/s10506-021-09306-3.

P. Rohera, C. Ginimav, G. Sawant, and R. Joshi. Better to ask in English? Evaluating factual accuracy of multilingual LLMs in English and low-resource languages. *arXiv:2504.20022*, 2025.

K. Trivedi, M. Shaikh, and S. Sharma. "So, how much do LLMs hallucinate on low-resource languages?" A quantitative and qualitative analysis. In *Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM)*, 2026. https://aclanthology.org/2026.loreslm-1.24/.

Z. Xu, S. Jain, and M. Kankanhalli. Hallucination is inevitable: An innate limitation of large language models. *arXiv:2401.11817*, 2024. https://arxiv.org/abs/2401.11817.

Labour Protection Act, B.E. 2541 (1998). *Royal Thai Government Gazette*, vol. 115, pt. 8a, p. 1, 20 February 1998. https://www.ratchakitcha.soc.go.th/DATA/PDF/2541/A/008/1.PDF.

Labour Protection Act (No. 7), B.E. 2562 (2019). *Royal Thai Government Gazette*, vol. 136, pt. 43a, p. 21, 5 April 2019. https://www.ratchakitcha.soc.go.th/DATA/PDF/2562/A/043/T_0021.PDF.

Emergency Decree Amending the Civil and Commercial Code, B.E. 2564 (2021). *Royal Thai Government Gazette*, vol. 138, pt. 26a, p. 1, 10 April 2021. https://www.ratchakitcha.soc.go.th/DATA/PDF/2564/A/026/T_0001.PDF.

Act Amending the Civil and Commercial Code (No. 23), B.E. 2565 (2022). *Royal Thai Government Gazette*, vol. 139, pt. 69a, p. 1, 8 November 2022. https://www.ratchakitcha.soc.go.th/DATA/PDF/2565/A/069/T_0001.PDF.

Act Amending the Civil and Commercial Code (No. 25), B.E. 2568 (2025). *Royal Thai Government Gazette*, vol. 142, pt. 14a, p. 1, 24 March 2025. https://ratchakitcha.soc.go.th/documents/58012.pdf.

Labour Protection Act (No. 9), B.E. 2568 (2025). *Royal Thai Government Gazette*, vol. 142, pt. 74a, p. 41, 7 November 2025. https://ratchakitcha.soc.go.th/documents/89818.pdf.

A Related work

Legal hallucination. Dahl et al. [2024] provide the first systematic profile of legal hallucination in public-facing models, reporting rates between 58% and 88% on verifiable questions about US federal cases, and that models often cannot predict when they are hallucinating. They also find performance strongest in the most frequently cited jurisdictions, which is the pattern one would expect to reappear, more sharply, in a jurisdiction whose statutes are written in a lower-resource language. We ask a different question: we hold the jurisdiction fixed and vary *time*.

Legal NLP in low-resource jurisdictions. Work here has concentrated on retrieval and citation rather than on what a model knows without retrieval. Atukorala et al. [2026] propose a resource-efficient retrieval blueprint for legal information in low-resource jurisdictions, and Akarajaradwong et al. [2025] introduce NitiBench, a Thai legal question-answering benchmark on which Thai legal QA systems are now commonly evaluated. Both measure performance when the relevant provisions are supplied; we measure the complementary case.

Reliability in low-resource languages. Trivedi et al. [2026] evaluate hallucination in Burmese and Armenian question answering and find that smaller monolingual models can appear to hallucinate less while producing vague answers — a caution we take seriously. Rohera et al. [2025] ask whether multilingual models are more accurate in English than in the user's language; our RQ4 asks the same for Thai law and does not find a consistent direction. Xu et al. [2024] argue that hallucination is an innate limitation rather than a solvable defect, which is part of why we treat retrieval rather than scale as the remedy.

What we measure. We follow Medvedeva et al. [2023] in distinguishing outcome forecasting from outcome identification. Our task is neither: we measure statute identification and application, and make no claim about how a court would decide. The closest recent system is OpenThai 2.0 Legal [iApp Technology, 2026], whose developers report closed-book recall of 0.07 on the Civil and Commercial Code and 0.99 open-book, and note that "coverage of niche areas or very recent amendments may be thinner."

B Rubric development, exclusions and reproducibility

Excluded items. A09 could be answered from arithmetic in the prompt without knowing the provision: the Thai-specific model scored 6/6 on it while answering incorrectly on 50 of its 60 responses in the labour control group. B10's answer is guessable from general M&A knowledge. C01 asks about conduct that falls outside parental authority under both the old and new wording, so it cannot separate the two eras; a model quoted the deleted phrase verbatim and still scored correct. These criteria were formed after inspecting pilot data.

Scorer bugs. Two classes recurred: patterns matching a topic rather than an answer (so a negated response still matched), and guard patterns not widened in step with the patterns they guard (so the most correct response in a set was scored as anchored). We caught the latter three times. Every revision is recorded in the released change log.

Reproducibility observations. At temperature 0, 113 of 962 model$\times$item$\times$language cells produced different labels across the three runs. Reasoning models were worse: one returned different labels on 35% of cells, and changing its reasoning-effort setting alone changed the legal answer from 3% to 5% — two different provisions of the same code. Some responses returned empty text, concentrated in one model on the longest multi-part questions. A single-run evaluation of a reasoning model on legal questions may not reproduce.

Grader validation detail. Agreement was perfect on the matched-pair item (48/48) and lowest on the straddle item requiring two dated answers in one response (72.2%); counting only positive mislabels, agreement is 92.3% [87.0–95.5].

C Full tables

C.1 Repealed-statute citation rate, all models

Table 4: RSCR by model and prompt language, post-frame items.

Model	Thai	English
claude-4.5	17.3% [10.4–27.4] (13/75)	40.0% [29.7–51.3] (30/75)
claude-5	10.3% [5.3–19.0] (8/78)	13.9% [8.0–23.2] (11/79)
gemini-2.5	20.8% [13.2–31.1] (16/77)	13.3% [7.4–22.8] (10/75)
gemini-3.1	17.1% [10.3–27.1] (13/76)	14.8% [8.7–24.1] (12/81)
gpt-4o	26.8% [17.9–38.1] (19/71)	23.1% [15.1–33.6] (18/78)
gpt-5.5	7.6% [3.5–15.6] (6/79)	14.8% [8.7–24.1] (12/81)
qwen3	30.7% [21.4–41.8] (23/75)	40.8% [30.4–52.0] (31/76)
typhoon-2.5	21.8% [14.1–32.2] (17/78)	27.0% [18.2–38.1] (20/74)
gpt-5*	4.9% [1.3–16.1] (2/41)	0.0% [0.0–8.8] (0/40)

C.2 Over-application of the current rule

Table 5: Rate at which the current rule is applied to facts predating the amendment, pre-frame items.

Model	Thai	English
claude-4.5	25.0% [13.8–41.1] (9/36)	10.0% [3.5–25.6] (3/30)
claude-5	0.0% [0.0–9.6] (0/36)	0.0% [0.0–9.6] (0/36)
gemini-2.5	2.9% [0.5–14.5] (1/35)	6.7% [1.8–21.3] (2/30)
gemini-3.1	0.0% [0.0–9.6] (0/36)	0.0% [0.0–10.4] (0/33)
gpt-4o	16.1% [7.1–32.6] (5/31)	27.6% [14.7–45.7] (8/29)
gpt-5.5	0.0% [0.0–9.6] (0/36)	8.3% [2.9–21.8] (3/36)
qwen3	18.8% [8.9–35.3] (6/32)	15.6% [6.9–31.8] (5/32)
typhoon-2.5	24.2% [12.8–41.0] (8/33)	37.9% [22.7–56.0] (11/29)

C.3 Anchoring by amendment age

Table 6: RSCR pooled across models, by amendment group. Ages relative to August 2026.

Group	Age	Thai	English
A (interest)	5 yr 4 mo	17.0% [12.8–22.3] (40/235)	21.7% [16.9–27.3] (52/240)
B (company promoters)	3 yr 6 mo	3.7% [1.6–8.4] (5/135)	10.4% [6.3–16.8] (14/134)
C (child punishment)	17 mo	28.3% [18.5–40.8] (17/60)	30.8% [19.9–44.3] (16/52)
D (maternity leave)	8 mo	25.0% [19.7–31.1] (55/220)	26.6% [21.3–32.6] (62/233)